

Current Scenario

Product images come from multiple vendors

No control over:

- Aspect ratios
- Backgrounds
- Framing
- Positioning

Impact on Business

- Product visibility issues (cropped or too small)
- Inconsistent UI → poor user experience
- Misaligned thumbnails → lower trust
- Reduced conversion rates



Low Product Coverage



Image not fitting in standardized container ratio, leading to white space BG



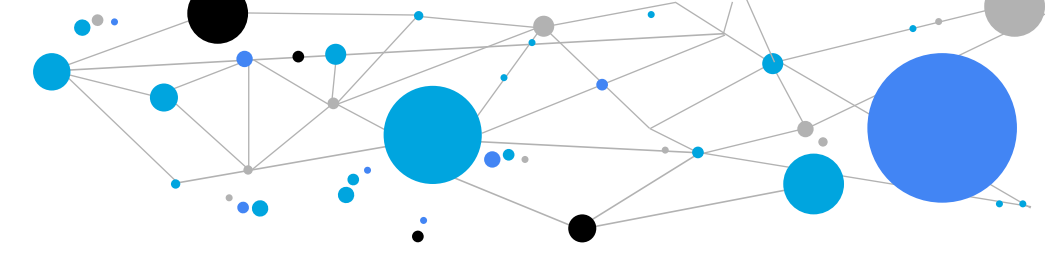
Cropped Product thumbnails



Model Crop



Product / Model not at centre



Proposed Solution Workflow: Product & BG Differentiation + BG Crop + BG Stretch

We use a two-stage segmentation pipeline : a domain-specific transformer-based model followed by a general-purpose foundation segmentation model, both producing pixel-level masks which are converted into bounding regions.

Stage	Condition (Trigger)	What Happens	Model / Method Used	Output / Action	Stop Condition	Base
Step 0: Aspect Ratio Identification	Image received	Extract aspect ratio from: • URL (preferred) • Image dimensions (fallback)	Regex + OpenCV	Compute W/H ratio	If ratio = 0.75 → STOP	
Step 1: Product Boundary detection + crop	Ratio ≠ 0.75	Detect product + background	Hugging Face Model (fashn-human-parser)	Generate segmentation mask → bounding box → attempt safe crop	1: Crop extent until Image reaches target aspect ratio OR 2. Until Product boundary saturates	BACKGROUND SUBTRACTION / CROP
Step 2 (Fallback) <i>HF model fails to produce a geometrically valid bounding region</i>	Product not in trained items of HF model & ratio!≠0.75	Detect product using general segmentation	SAM (Segment Anything – Meta)	Generate masks → filter → union → bounding box → safe crop	Same as above	BACKGROUND SUBTRACTION / CROP

Product
List

Original
<i>W/H (original image)</i>
1.0000



STEP 1

BORDER DETECTION
&
BG CROP



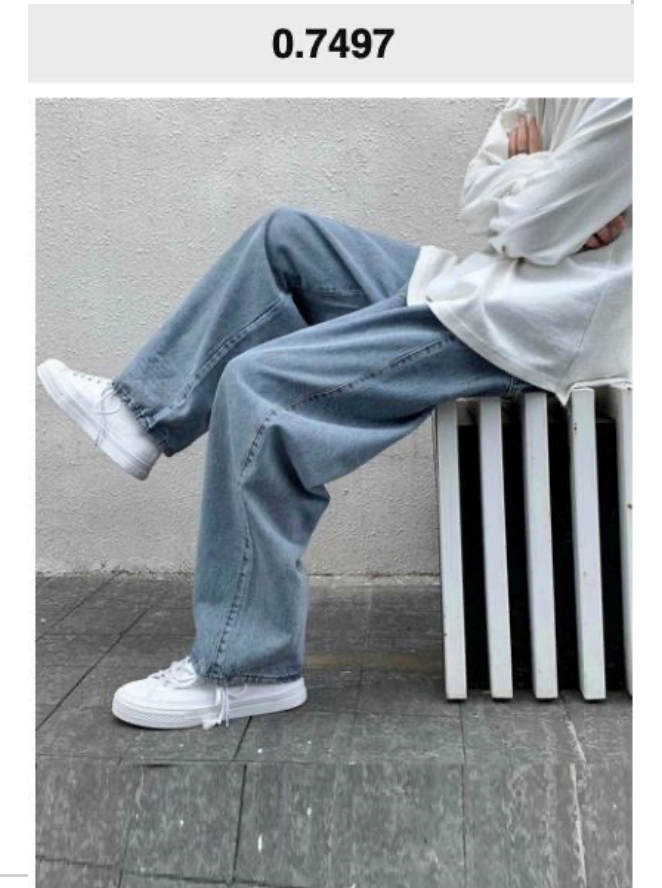
STEP 2

NA



STEP 3

BG STRETCH



Stage	Condition (Trigger)	What Happens	Model / Method Used	Output / Action	Stop Condition	Base
Step 3 - BG STRETCH	Ratio still $\neq$ 0.75	Adjust background only	Variance-based BG classification + OpenCV resize	Stretch background (horizontal / vertical)	Until Ratio $\approx$ 0.75 for solid/gradient image.. (Low var) Limited stretch for complex BGs (25% of BG Width)	BACKGROUND ADDITION / PADDING
Final Step	After all transformations	Recenter Product Box at mid	Plan: recentering of product bounding (not yet executed)	Final standardized image	END	IMAGE POSITIONING

Original
<i>W/H (original image)</i>
1.0000



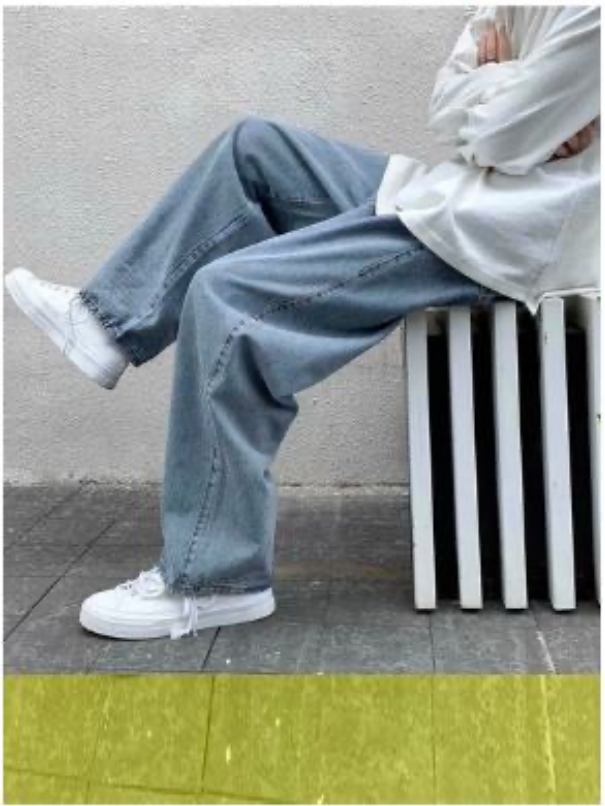
STEP 1
BORDER DETECTION
&
BG CROP

0.8950



STEP 2
NA

0.8950



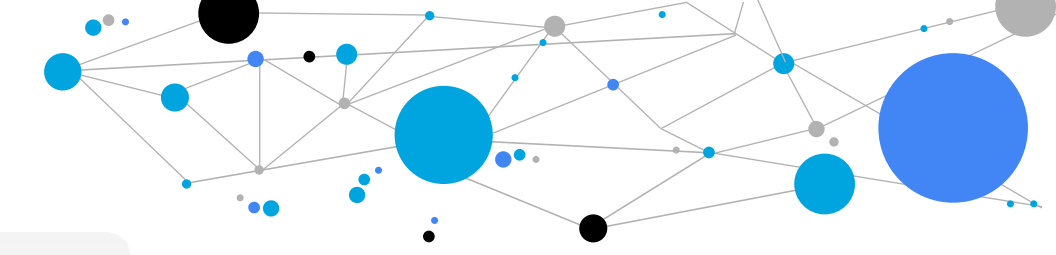
STEP 3
BG STRETCH

0.7497



Hard Constraints: - Product mask (Object Boundary) treated as immutable region
- All transformations applied only on background pixels

Slikk Aspect Ratio Fix



MVP Code Flaws

1. Float Error: Aspect Ratio reaches near 0.75 (eg: 0.7495/0.7505) and pipeline stops
2. SAM Model not working - which was designed as backup for product Fashn Parser not trained
3. BG Stretch - Replicating and stretching replicated BG in some cases. Expected: stretching pixel across BG, no BG replication. Works for solid, gradient, textured BG only

1.0000



0.7500



Model not trained to identify Bangles



Model recognized all image as BG, and stretched all image to fit from 1 to 0.75



Img ratio = 1

Wrong Detection saved by:

Crop Condition: 1: Crop extent until Image reaches target aspect ratio OR 2. Until Product boundary saturates
GOAL: Crop Image to achieve aspect ratio = 0.75, not achieve maximum product coverage

Additional Conditions required to improve in MVP:

1. 1% Gap b/w Product Border and Container to avoid product edge touching container, that feels like cropped product
2. Signature, Text, Tags inclusion in product boundary
3. Recenter product if bg available



Slikk Aspect Ratio Fix

MVP Model Fixes ~97% Images

Analysis 